AUTONOMOUS WAREHOUSE ROBOT

A PROJECT REPORT PHASE- I

submitted by

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ABSTRACT

KEYWORDS: Autonomous Warehouse Robot; Autonomous Mobile Robot; HuskyLens; AprilTag; Obstacle Detection; Safety Monitoring; Arduino UNO R4 WiFi.

The rapid growth of automation and intelligent technologies in the manufacturing and logistics sectors has transformed conventional warehouse operations by improving efficiency, accuracy and safety. However, traditional warehouse material handling and inventory management still involve significant manual intervention, which can lead to errors, increased operational time and safety risks. With the advancement of Industry 4.0 technologies, autonomous mobile robots have emerged as a promising solution for automating warehouse operations through intelligent navigation, object detection, identification and real-time monitoring.

This project focuses on the development of an **Autonomous Warehouse Robot** designed for material handling, inventory identification, obstacle detection and safety monitoring in warehouse environments. The proposed system integrates AI-based vision, sensors, embedded control, and wireless communication to enable autonomous operation. The robot uses **HuskyLens for object and obstacle detection**, while an **AprilTag-based identification system** is incorporated for identifying designated warehouse locations or items. A servo mechanism provides 180° scanning of the surroundings, while an Arduino UNO R4 WiFi processes the sensor information and controls the movement of the robot through DC geared motors and IBT-2 motor drivers.

The system also incorporates gas and flame detection to improve operational safety. When an obstacle or hazardous condition is detected, the robot can respond appropriately by stopping or controlling its movement and generating alerts through an OLED display, buzzer, and Telegram-based remote notification. The proposed system aims to combine autonomous navigation, vision-based detection, identification, safety monitoring, and remote communication into a low-cost and practical warehouse automation platform. Overall, the project demonstrates the potential of intelligent mobile robots to reduce manual intervention, improve warehouse safety, and support efficient material handling and monitoring in modern warehouse environments.

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ABBREVIATIONS

AMR	Autonomous Mobile Robot
AI	Artificial Intelligence
DC	Direct Current
I/O	Input / Output
OLED	Organic Light-Emitting Diode
Wi-Fi	Wireless Networking Technology
IBT-2	BTS7960-Based Motor Driver Module
DC	Direct Current

CHAPTER 1

INTRODUCTION

1.1 Warehouse automation

Warehouses are an important part of manufacturing and distribution systems, where materials and products are received, stored, moved and dispatched. Traditional warehouse operations often depend on manual handling and transportation of materials. As the size and complexity of warehouse operations increase, manual handling can become time-consuming and may lead to inconsistencies in material movement and identification.

Warehouse automation involves the use of mechanical, electrical and computer-based systems to improve the efficiency of warehouse operations. Automated storage systems, conveyors, automated guided vehicles and autonomous mobile robots are commonly used for different warehouse activities. Among these, mobile robots provide greater flexibility because they can move between different locations instead of being restricted to a fixed position.

Autonomous mobile robots can perform functions such as navigation, object detection, obstacle avoidance and location identification with limited human intervention. The integration of sensors, controllers and actuators allows such robots to sense their surroundings and respond according to the detected conditions.

The present project focuses on the development of an Autonomous Warehouse Robot prototype. The main focus of the project is on navigation, identification, detection and monitoring functions required for autonomous operation in a warehouse-like environment.

1.2 Autonomous mobile robots in warehouses

An Autonomous Mobile Robot (AMR) is a mobile robotic system capable of sensing its surroundings and making decisions related to its movement without continuous manual control. A typical autonomous mobile robot consists of a controller, sensing system, drive mechanism and power supply. The controller processes the information received from the sensors and generates suitable commands for the actuators.

In warehouse applications, autonomous mobile robots can be used for transportation, navigation, monitoring and identification of different locations. Their ability to

move independently makes them suitable for structured environments where predefined paths or checkpoints can be established.

Navigation and obstacle detection are important functions of an autonomous warehouse robot. The robot must be able to follow its intended path while identifying objects or obstacles in its surroundings. In the proposed system, visual sensing is used to support these functions. A line-following approach is used for navigation, while the vision system is also used for object and obstacle detection.

The proposed robot uses an Arduino UNO R4 WiFi as the main controller and 12 V DC geared motors for movement. The motors are controlled using IBT-2 motor drivers. A DFRobot HuskyLens is used as the primary vision unit for line following, object detection, obstacle detection and AprilTag identification.

1.3 Vision-based navigation and identification

Vision-based sensing allows a robot to obtain information about its surroundings using visual features. It can be used for navigation, object detection and identification of specific locations. In the proposed system, the DFRobot HuskyLens is used to provide the required visual information.

For navigation, the HuskyLens is used for line following. The detected line provides a reference for the robot's movement, allowing the controller to generate appropriate motor commands. The same vision system is also used for detecting objects and obstacles in the robot's operating area.

AprilTags are used for visual identification of selected locations or checkpoints. The robot can detect an AprilTag using the HuskyLens and use the detected information as part of its identification and navigation process. This provides a simple method for identifying predefined locations in a structured warehouse environment.

The HuskyLens is mounted on a servo mechanism that allows it to rotate through approximately 180°. This provides a wider viewing range and allows the robot to scan different directions while operating. The combination of line following, object detection, obstacle detection, AprilTag identification and servo-based scanning forms the main vision-based functionality of the proposed robot.

1.4 Need for autonomous warehouse robots

The increasing requirement for efficient material handling and monitoring has created a need for flexible automation systems in warehouses. Manual transportation and monitoring can require continuous human effort, particularly when the same operations have

to be repeated frequently. Autonomous mobile robots can assist in such environments by performing movement and monitoring functions with reduced human intervention.

A warehouse robot should be capable of navigating through its operating area while identifying obstacles and important locations. The use of visual identification can help the robot recognize predefined checkpoints, while obstacle detection allows it to respond to changes in its path.

The proposed project provides a platform for studying the integration of mechanical fabrication, embedded control, motor actuation and machine vision. The project mainly concentrates on navigation, identification, detection and monitoring, rather than material manipulation.

Safety monitoring is also considered as part of the future development of the system. Gas and flame sensors are planned to be integrated for detecting potentially unsafe conditions. Similarly, OLED, buzzer and Telegram-based monitoring are considered for providing local and remote status information.

1.5 Overview of proposed system

The proposed Autonomous Warehouse Robot consists of a mobile chassis, drive system, controller, vision system and supporting monitoring components. The Arduino UNO R4 WiFi acts as the main controller and coordinates the operation of the different subsystems.

The DFRobot HuskyLens provides the primary visual input for line following, object detection, obstacle detection and AprilTag identification. A servo motor is used to rotate the HuskyLens through approximately 180° for environmental scanning. The robot is driven using 12 V DC geared motors connected through IBT-2 motor drivers.

The system is powered using a 12 V, 6000 mAh battery. Additional components such as an OLED display and buzzer are intended for local indication, while Telegram-based monitoring is planned for remote status and alert communication. Gas and flame sensors are also planned as part of the future safety monitoring system.

The overall operation of the proposed system can be represented by the following sequence:

Sensing $\rightarrow$ Processing $\rightarrow$ Decision Making $\rightarrow$ Movement and Scanning $\rightarrow$ Monitoring

The present Phase-I work includes mechanical fabrication and testing of motor movement, HuskyLens detection, AprilTag identification and servo-based scanning.

The individual functions are being tested as part of the prototype development. Further integration, safety sensing and monitoring features will be developed and tested in the subsequent phase.



Figure 1.1: Block diagram of the proposed Autonomous Warehouse Robot

Thus, the project aims to develop a functional prototype that demonstrates the basic capabilities required for an autonomous warehouse robot, with emphasis on navigation, visual identification, obstacle detection and monitoring.

CHAPTER 2

LITERATURE REVIEW

The development of autonomous warehouse robots involves several areas of robotics, including autonomous navigation, perception, obstacle avoidance, visual identification and mobile robot control. A review of existing research helps to understand the technologies used in autonomous mobile robots and the challenges associated with their application in warehouse environments. The following studies were considered in relation to the proposed Autonomous Warehouse Robot.

2.1 Giuseppe fragapane, rené de koster, fabio sgarbossa and jan ola strandhagen: "planning and control of autonomous mobile robots for intralogistics: literature review and research agenda" (2021)

Fragapane et al. presented a detailed review of autonomous mobile robots used in intralogistics applications such as manufacturing, warehousing, cross-docking and terminals. The study discussed the development of autonomous mobile robots from conventional automated guided vehicles and examined the effect of advances in hardware and control systems on robot planning and operation.

The authors highlighted that autonomous mobile robots provide greater flexibility in dynamic environments because they can make decisions based on changes in their surroundings. The review covered important aspects such as navigation, routing, scheduling and control of autonomous mobile robots. The study also identified future research requirements related to the planning and control of autonomous mobile robots.

This work provides a general foundation for the present project by demonstrating the importance of autonomous navigation and control in warehouse and intralogistics applications. The proposed robot similarly focuses on autonomous movement and environmental response, but is developed as a small-scale prototype with emphasis on vision-based detection and identification.

2.2 Edwin olson: "AprilTag: a robust and flexible visual fiducial system" (2011)

Edwin Olson introduced AprilTag, a visual fiducial system designed for robotic perception and identification. The system uses coded visual tags that can be detected from images and provides information about the location and orientation of the tag. The

study described the use of artificial visual features for applications where controlled and identifiable reference points are required.

The proposed AprilTag system was designed to provide robust detection and showed improved performance in situations involving occlusion, image distortion and changes in viewing conditions. The work demonstrated the usefulness of visual fiducial markers for robot navigation and perception.

AprilTags are relevant to the present project because they provide a simple method for identifying predefined locations or checkpoints. In the proposed Autonomous Warehouse Robot, AprilTag identification is implemented using the HuskyLens. The detected tags can be used as visual references during navigation and location identification.

2.3 Sven franke, rené de koster, christopher reining, havva celik and alice kirchheim et al.: "automated and autonomous mobile robots in warehousing: definitions, autonomy levels, and research directions" (2026)

Franke et al. presented a systematic review of automated and autonomous mobile robots used in warehousing. The study examined the different ways in which terms such as Automated Guided Vehicle (AGV), Autonomous Mobile Robot (AMR) and Mobile Robot are used in existing research. The authors analysed research published between 2000 and 2025 and classified mobile robotic systems according to characteristics such as control, manipulation, collaboration and software capabilities.

The study highlights the increasing importance of autonomous and mobile robotic systems in warehouse applications. It also indicates that the level of autonomy of a robotic system depends on its sensing, decision-making and operational capabilities.

The proposed project is a prototype-level autonomous mobile robot that integrates sensing, control and movement functions. Although it does not aim to provide the complete level of autonomy found in industrial warehouse systems, the project applies the basic principles of autonomous operation through visual sensing, navigation, obstacle detection and location identification.

2.4 Chetan zaware, r. s. gadakh, urmila s. nagargoje, mayur j. gitay, balaprasad p. kurpatwar and amol p. ghule: "autonomous mobile robots in warehouse automation: a comprehensive review of AI, navigation, and sensing technologies under industry 4.0" (2026)

Zaware et al. reviewed the application of autonomous mobile robots in warehouse automation with emphasis on artificial intelligence, navigation and sensing technologies. The review considered areas such as autonomous navigation, computer vision, object detection, path planning, obstacle avoidance, sensor fusion and intelligent warehouse operations.

The study shows that perception and navigation are important components of autonomous warehouse robots. The ability of a robot to identify objects and obstacles and respond appropriately is essential for reliable operation in a warehouse environment.

The present project applies these concepts at a prototype level. The HuskyLens is used for visual perception, including line following, object detection and obstacle detection. AprilTag identification is also incorporated for recognizing selected locations. These functions are combined with motor control to develop the mobile robotic platform.

2.5 Research gap

The literature shows that autonomous mobile robots have been extensively studied for warehouse and intralogistics applications. Existing research covers advanced navigation, path planning, perception, obstacle avoidance, fleet coordination and intelligent decision-making. AprilTags have also been established as useful visual fiducial markers for robotic perception and navigation.

However, many advanced warehouse robotic systems require complex sensing, computing and navigation frameworks. There is therefore a need for simpler prototype systems that can demonstrate the fundamental functions of an autonomous warehouse robot using comparatively accessible hardware.

The present project addresses this area by developing a compact prototype that combines line following, object detection, obstacle detection, AprilTag identification and servo-based visual scanning. The system uses an Arduino UNO R4 WiFi as the main controller and a HuskyLens as the primary vision unit. The project focuses on integrating these individual functions into a practical mobile robotic platform suitable for studying the basic principles of autonomous warehouse operation.

CHAPTER 3

OBJECTIVES

The main objective of the project is to develop a prototype of an Autonomous Warehouse Robot capable of performing basic navigation, identification, detection and monitoring functions with minimal human intervention. The system is designed by integrating a mobile platform, embedded controller, vision-based sensing and motor control. The specific objectives of the project are as follows.

3.1 Autonomous navigation

To develop a mobile robotic platform capable of autonomous movement along a predefined path using vision-based line following. The system is intended to maintain its movement along the designated path and provide controlled motion using DC geared motors.

3.2 Object and obstacle detection

To implement vision-based detection of objects and obstacles in the operating environment using the DFRobot HuskyLens. The detected information is intended to assist the robot in understanding its surroundings during autonomous movement.

3.3 Obstacle avoidance

To develop an obstacle avoidance mechanism that enables the robot to respond when an obstacle is detected in its path. The system should be capable of modifying its movement or stopping as required to avoid collisions during operation.

3.4 AprilTag-based identification

To implement AprilTag identification for recognizing predefined locations or checkpoints in the warehouse environment. The HuskyLens is used to detect the AprilTags, allowing the robot to obtain visual information about specific locations during its operation.

3.5 180° Environmental Scanning

To develop a servo-based scanning mechanism for rotating the HuskyLens through approximately 180°. This allows the vision system to observe different directions around the robot and improves its ability to detect objects and obstacles within the surrounding area.

3.6 Monitoring and alert system

To provide provisions for monitoring the operating condition of the robot using an OLED display and buzzer. The project also considers Telegram-based remote monitoring for transmitting selected status or alert information during operation.

3.7 Safety monitoring

To provide provision for future integration of gas and flame sensors for detecting potentially unsafe conditions in the operating environment. These sensors are planned as part of the subsequent development and will be tested after integration with the main system.

3.8 Development of an integrated prototype

To integrate the mechanical, electrical and control components into a functional autonomous warehouse robot prototype. The Phase-I work focuses on mechanical fabrication and testing of motor movement, HuskyLens detection, AprilTag identification and servo-based scanning, providing a foundation for further system integration and testing in the subsequent phase.

CHAPTER 4

PROJECT PLAN AND TASK ALLOCATION

The project is divided into two phases. Phase 1 focuses on the study, development and testing of the basic prototype functions, while Phase 2 focuses on further integration, improvement and testing of the complete system.

4.1 Phase 1

Phase 1 deals with the study of autonomous warehouse robots and the development of the basic prototype. The initial work includes understanding the requirements of the system, studying relevant literature, selecting suitable components and developing the mechanical structure of the robot.

The prototype is developed by integrating the Arduino UNO R4 WiFi, HuskyLens, DC geared motors, IBT-2 motor drivers, servo motor and other supporting components. The mechanical structure is fabricated to provide a suitable platform for mounting the different components.

The basic functions of the robot are tested during this phase. These include motor movement, HuskyLens-based detection, AprilTag identification, line following and servo-based scanning. The testing of these individual functions helps in identifying problems and making the required improvements before complete system integration.

As per the current scope, the Phase-I project plan and progress are shown in Figure 4.1.

PROJECT PLAN FOR PHASE I

Duration of week **Name of task**	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Literature survey								
Component selection and study								
Mechanical fabrication								
Motor and drive system testing								
HuskyLens and AprilTag testing								
Line following and servo scanning								
Safety sensor and alert integration								
Report writing and review								

Figure 4.1: Project Plan for Phase I

Table 4.1: Task Allocation

Sl. No.	Area of Work	Major Tasks	Team Member
1	Mechanical Development	Chassis development, component mounting and mechanical fabrication	A S Shivani
2	Electronics and Motor Control	Motor driver connections, power system and motor movement	Abeesha K Shaji
3	Vision and Identification	HuskyLens integration, object detection and AprilTag identification	Abey Thomas
4	Programming and Integration	Controller programming, servo scanning and system integration	Anjana S
5	Testing and Documentation	Functional testing, result recording and report preparation	All Team Members

Note: The final task allocation can be modified according to the actual work carried out by each team member.

4.2 Phase 2

Phase 2 deals with the further integration and testing of the prototype. The individual functions developed during Phase 1 will be combined to achieve more reliable autonomous operation of the robot.

The major activities of Phase 2 include improvement of navigation, obstacle avoidance and AprilTag-based identification. The monitoring and alert features using the OLED display and buzzer will be integrated and tested. The planned gas and flame sensors will also be incorporated for safety monitoring.

Further testing will be carried out under different operating conditions to evaluate the response of the robot to obstacles, visual markers and changes in its path. Telegram-based remote monitoring will also be developed and tested as part of the future monitoring system.

The final stage of Phase 2 will involve overall system testing and evaluation. The performance of the integrated prototype will be analysed based on navigation, detection, identification, scanning and monitoring functions. The results obtained will be used to identify possible improvements and future development areas.

CHAPTER 5

DESIGN AND METHODOLOGY

5.1 Design

The design of the Autonomous Warehouse Robot consists of a mobile mechanical structure, drive system, sensing system, controller and supporting components. The different subsystems are integrated to develop a prototype capable of autonomous movement, visual detection, location identification and environmental scanning.

The robot is designed as a compact mobile platform with wheels and DC geared motors for movement. The electronic components are mounted on the fabricated structure, while the HuskyLens is positioned on a servo mechanism to provide a wider field of view. The Arduino UNO R4 WiFi is used as the main controller for coordinating the different functions of the robot.

5.1.1 Mechanical structure

The mechanical structure provides the base for mounting the motors, wheels, battery, controller and other components. The fabricated chassis supports the different subsystems and allows the robot to move on the floor.

The drive system consists of 12 V DC geared motors connected to the wheels. The motors provide the required torque for movement of the robot. The arrangement of the wheels and motors allows the robot to perform forward, reverse and turning movements.

The HuskyLens is mounted on a servo mechanism. The servo allows the vision unit to rotate through approximately 180°, providing a wider scanning range. This arrangement allows the robot to observe different directions while moving through its operating environment.

5.1.2 Hardware design

The Arduino UNO R4 WiFi acts as the main control unit of the robot. It receives information from the HuskyLens and controls the movement and scanning mechanisms according to the programmed logic.

The DC geared motors are controlled using IBT-2 motor drivers. The motor drivers provide the required current handling capability for operating the motors from the 12 V power supply. A servo motor is connected to the controller to rotate the HuskyLens for environmental scanning.

The robot is powered using a 12 V, 6000 mAh battery. A buck converter is used where required to obtain suitable supply levels for the electronic components. Supporting components include an OLED display, buzzer, switch and dot board.

Gas and flame sensors are planned for future safety monitoring and are not considered part of the completed Phase-I testing.

Table 5.1: Components used in the Autonomous Warehouse Robot

COMPONENTS	**QUANTITY**
Arduino UNO EK R4 Wifi	1
DF Robot Husky Lens	1
Servo Motor	1
DC Gear Motor 12V	4
Motor Driver- IBT-2	2
Flame Sensor	1
Gas Sensor	1
Chassis (Frame & Wheels)	
Battery 12V, 6000mAh	1
DotBoard	3
Buzzer	1
OLED Display	1
Buck Converter	1
Switch	1

5.1.3 System architecture

The overall system consists of four main sections: sensing, processing, actuation and monitoring.

The HuskyLens forms the main sensing unit and is used for line following, object detection, obstacle detection and AprilTag identification. The Arduino UNO R4 WiFi processes the received information and coordinates the operation of the robot.

The movement system consists of the IBT-2 motor drivers and 12 V DC geared motors. The servo motor controls the viewing direction of the HuskyLens. OLED

and buzzer-based indication are included as supporting monitoring components, while Telegram-based remote monitoring is planned for future implementation.

The basic system flow is:

HuskyLens / Sensors → Arduino UNO R4 WiFi → Decision Making → Motor and Servo Control → Monitoring

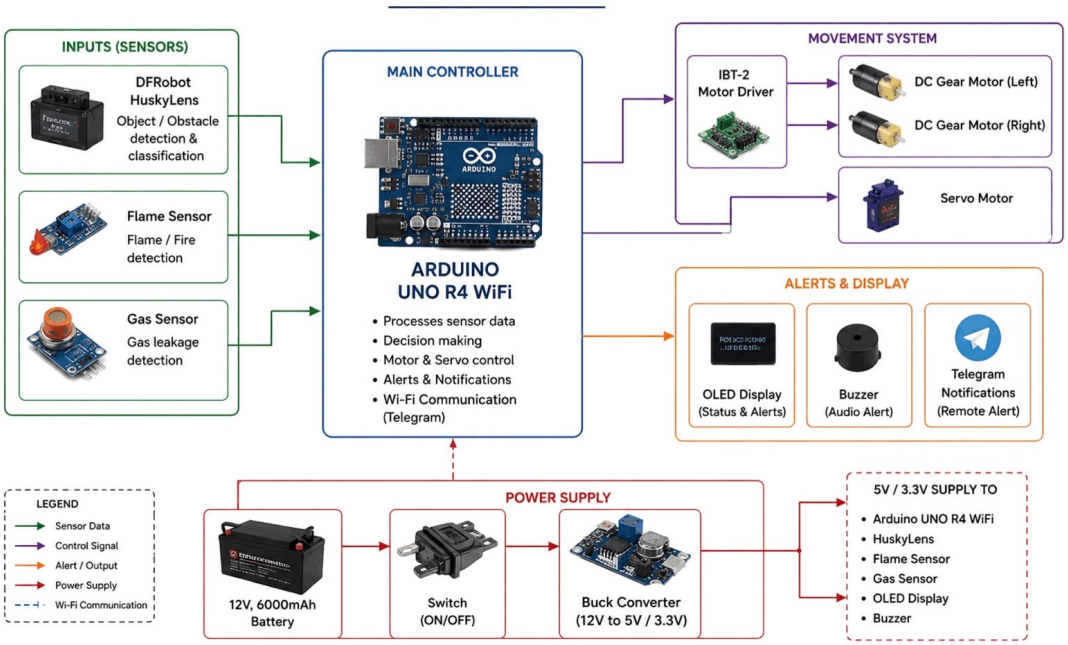


Figure 5.1: System Architecture of the Autonomous Warehouse Robot

5.2 Methodology

The methodology of the proposed Autonomous Warehouse Robot is based on the integration of vision-based sensing, embedded control and motor actuation. The robot continuously receives information from the vision system and uses the information to perform the required movement or identification function.

5.2.1 Line following

Line following is used as the basic navigation method for the robot. The HuskyLens detects the predefined path or line on the floor. The detected line position is used by the controller to determine the required direction of movement.

Based on the detected position of the line, suitable commands are provided to the motor drivers. The motors are operated accordingly to maintain the robot along the intended path.

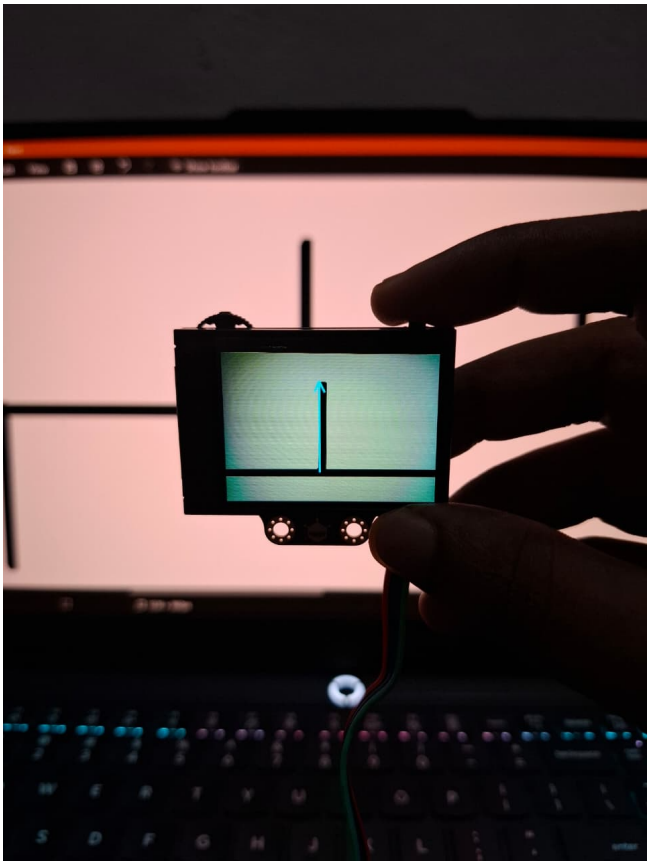


Figure 5.2: Line-following operation using HuskyLens

5.2.2 Object and obstacle detection

The HuskyLens is used to detect objects and obstacles present in the robot's operating environment. During autonomous movement, the vision system provides information about objects within its field of view.

When an obstacle is detected, the information is processed by the controller. The robot can then stop or modify its movement depending on the programmed response. This function is important for reducing the possibility of collisions during autonomous operation.

5.2.3 Obstacle avoidance

Obstacle avoidance is performed based on the information obtained from the detection system. When an obstacle is identified in the robot's path, the controller determines the appropriate movement response.

The robot may stop its forward motion and change its direction depending on the obstacle and the programmed control sequence. Further testing and improvement of the

obstacle avoidance sequence will be carried out during Phase 2.

5.2.4 AprilTag identification

AprilTags are used as visual markers for identifying predefined locations or checkpoints. The HuskyLens detects the AprilTag and provides the corresponding identification information to the controller.

The identified tag can be associated with a particular location or operation in the warehouse environment. This provides the robot with a simple visual reference for location identification.

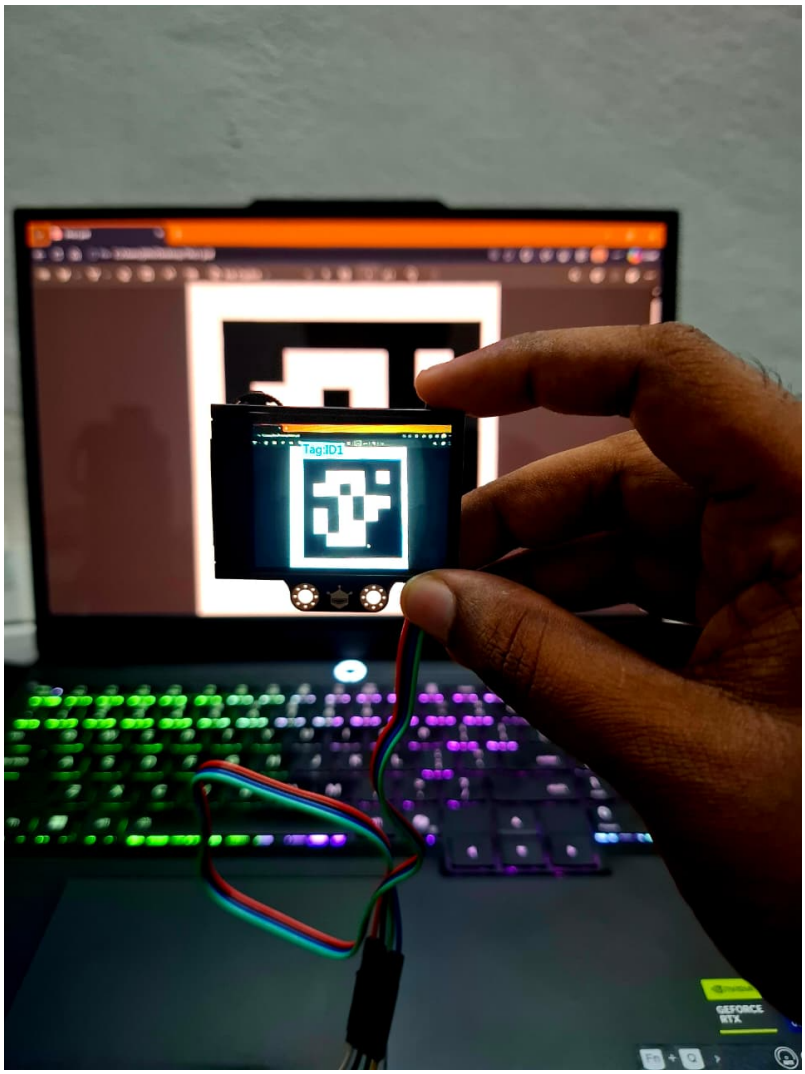


Figure 5.3: AprilTag identification using HuskyLens

5.2.5 Motor control

The movement of the robot is achieved using 12 V DC geared motors. The motors are connected to IBT-2 motor drivers, which are controlled by the Arduino UNO R4 WiFi.

The controller generates the required signals for the motor drivers based on the navigation and detection conditions. The motor system allows the robot to move forward, reverse and change direction during operation.

Initial motor movement testing has been carried out to verify the basic movement of the fabricated platform.

5.2.6 Monitoring and alerts

The proposed system includes provisions for local monitoring using an OLED display and buzzer. The OLED can be used to display selected system information, while the buzzer can provide an audible indication during specific conditions.

Telegram-based remote monitoring is also planned for future development. It can be used to transmit selected status or alert information to a remote user.

Gas and flame sensors are planned to be integrated as part of the safety monitoring system. Their implementation and testing will be carried out during the subsequent phase.

5.3 System working principle

This project is an Autonomous Warehouse Robot designed to monitor safety and navigate automatically inside a warehouse environment. The system mainly works in a continuous loop of sensing, processing, decision-making, and action.

First, the input section includes three main sensors. The HuskyLens AI camera is used for object and obstacle detection, so the robot can identify paths or avoid obstacles. A flame sensor detects fire, and a gas sensor detects gas leakage in the environment. These sensors send information to the main controller.

The Arduino UNO R4 WiFi is the main controller. It processes the sensor data and makes decisions based on programmed logic. For example, if an obstacle is detected, the robot can stop or change direction. If gas or fire is detected, the system triggers alerts. The controller also handles communication using Wi-Fi.

Based on these decisions, the movement system is activated. IBT-2 motor drivers

control the DC geared motors for movement, allowing the robot to move forward, backward, or turn. A servo motor rotates the HuskyLens for directional scanning.

For user feedback and safety alerts, an OLED display shows status messages, a buzzer provides immediate audio alerts, and Telegram notifications support remote monitoring. The system is powered by a 12 V battery through a switch and a buck converter, which provides suitable lower voltages for the controller and other electronic components.

Overall, the robot continuously monitors its surroundings, processes sensor information, performs the programmed response and alerts the user when a potential hazard is detected.

Start $\rightarrow$ System Initialization $\rightarrow$ Vision Detection $\rightarrow$ Decision Making $\rightarrow$ Navigation / Identification / Obstacle Response $\rightarrow$ Scanning and Monitoring $\rightarrow$ Continue Operation

CHAPTER 6

PRELIMINARY RESULTS

The preliminary results of the project are based on the work completed during Phase-I of the Autonomous Warehouse Robot. The initial phase included the study of autonomous warehouse robots, literature review, identification of project requirements, selection of hardware components and development of the prototype. The major components were integrated and individual functions were tested to evaluate their suitability for the proposed system.

Some of the functions, such as HuskyLens integration, AprilTag identification and servo-based scanning, have been completed successfully. Mechanical fabrication, motor and drive system implementation, motor movement testing, object and obstacle detection testing and line-following implementation are currently in progress. The remaining monitoring, safety and complete system testing activities are planned for the subsequent phase.

6.1 Mechanical structure development and fabrication

The mechanical structure of the Autonomous Warehouse Robot is currently under development and fabrication. The structure provides the base for mounting the drive motors, wheels, battery, controller and other electronic components.

The fabrication work is being carried out according to the requirements of the proposed prototype. The mechanical structure is being developed to provide sufficient stability for the movement and mounting of the sensing system.

The mechanical development is currently in progress, and further modifications may be carried out during the subsequent stages of the project.

6.2 Motor and drive system

The motor and drive system is currently being implemented using DC geared motors and IBT-2 motor drivers. The drive system is responsible for providing movement to the robot.

The motors and motor drivers have been integrated with the prototype, and the movement system is currently undergoing testing. Further testing is required to verify

reliable movement and proper response of the drive system under different operating conditions.

6.3 HuskyLens integration

The DFRobot HuskyLens has been successfully integrated into the prototype as the primary vision unit. It is intended to provide the visual information required for navigation, object detection, obstacle detection and AprilTag identification.

The integration of the HuskyLens with the prototype has been completed. The vision unit is mounted on a servo mechanism so that its viewing direction can be changed during operation.

6.4 AprilTag identification testing

AprilTag identification has been implemented and tested using the HuskyLens. The testing demonstrated the detection and identification of the selected AprilTag marker.

AprilTags can be used as visual references for identifying predefined locations or checkpoints in the proposed warehouse environment. Further testing can be carried out under different distances and orientations during the subsequent development of the system.

6.5 Servo-based HuskyLens scanning

A servo motor is used to rotate the HuskyLens for environmental scanning. The servo-based scanning mechanism has been implemented and tested to provide approximately 180° movement of the vision unit.

The scanning mechanism allows the HuskyLens to observe different directions around the robot instead of being restricted to a fixed viewing direction. This function is intended to support the detection and monitoring capabilities of the robot.

6.6 Motor movement testing

Motor movement testing is currently in progress. The DC geared motors and IBT-2 motor drivers are being tested to obtain controlled movement of the robot.

The testing is being carried out to verify the basic movement of the robot and to identify any required modifications in the motor control and mechanical arrangement. Further testing will be required before integrating the complete navigation sequence.

6.7 Object and obstacle detection testing

Object and obstacle detection using the HuskyLens is currently under testing. The purpose of this testing is to evaluate the ability of the vision system to detect objects and obstacles present in the operating environment.

The information obtained from the detection system will be used for developing the obstacle response of the robot. Further testing and refinement are required before the complete obstacle avoidance function can be considered.

6.8 Line-following implementation and testing

Line-following implementation and testing are currently in progress. The HuskyLens is being used to detect the predefined path, and the detected information will be used to control the movement of the robot.

The line-following function is an important part of the navigation system. Further testing will be carried out to improve the movement response and maintain reliable alignment with the predefined path.

6.9 Overall phase-i status

The current status of the major activities carried out during Phase-I is summarized in Table 6.1.

Table 6.1: Status of Phase-I Activities

Sl. No.	Activity	Status
1	Study of autonomous warehouse robots and warehouse automation	Completed
2	Literature review	Completed
3	Identification of project requirements	Completed
4	Selection of hardware components	Completed
5	Mechanical structure development and fabrication	In Progress
6	Motor and drive system implementation	In Progress
7	Motor movement testing	In Progress
8	HuskyLens integration	Completed
9	Object and obstacle detection testing	In Progress
10	AprilTag identification testing	Completed
11	Servo-based HuskyLens scanning	Completed
12	Line-following implementation and testing	In Progress
13	Integration of monitoring and alert components	Planned
14	Gas and flame sensor integration	Planned
15	Overall system testing	Planned for Phase 2

The Phase-I work has established the basic foundation for the development of the Autonomous Warehouse Robot. The study and component-selection activities have been completed, while important prototype functions such as HuskyLens integration, AprilTag identification and servo-based scanning have been implemented.

The mechanical structure, motor and drive system, motor movement, object and obstacle detection and line-following functions are still under development and testing. Monitoring and alert components, gas and flame sensor integration and complete system testing will be addressed in the subsequent phase.

Thus, the preliminary results indicate that the major building blocks required for the proposed system have been initiated, while further integration and testing are necessary to obtain a fully functional autonomous warehouse robot.

CHAPTER 7

CONCLUSION

The Phase-I of the project focused on the study, development and preliminary testing of an Autonomous Warehouse Robot for navigation, identification, object and obstacle detection, and monitoring applications in a warehouse environment. A detailed study of autonomous warehouse robots and warehouse automation was carried out, followed by a literature review to understand the existing technologies and approaches used in autonomous mobile robots.

Based on the identified requirements, the major hardware components were selected and the initial prototype development was started. The Arduino UNO R4 WiFi was selected as the main controller, while HuskyLens was integrated for vision-based functions such as line following, object detection, obstacle detection and AprilTag identification. A servo motor was incorporated to provide approximately 180° scanning of the surrounding environment. The motor and drive system using 12 V DC geared motors and IBT-2 motor drivers was also initiated.

During Phase-I, HuskyLens integration, AprilTag identification testing and servo-based scanning were completed. Mechanical fabrication, motor movement testing, object and obstacle detection testing, and line-following implementation are currently in progress. The integration of monitoring and alert components, gas and flame sensors, and complete system testing are planned for the next phase.

Thus, Phase-I has established the basic foundation for the development of the Autonomous Warehouse Robot. The work completed so far provides a base for further integration and testing in Phase-II, with emphasis on improving navigation, obstacle avoidance, identification, monitoring and overall system performance.

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